॥वसुधैव कुटुम्बकम्॥

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE - Academic Year 2026-27

Java Programming – Lab (Batch 2025-29)

Lab Assignment No:- 30

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Batch	2025-29
Class	Programming with JAVA
Academic Year & Semester	2nd Year – Sem III
Title of Assignment	Develop a Java application to demonstrate CRUD operations.
Practice questions	1. Create a JSP-based application to add, update, delete and view employee details.

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8" %>
<%@ page import="java.sql.*" %>
<%!
    // Database settings - change the password to match your MySQL
setup
    final String DB_URL = "jdbc:mysql://localhost:3306/testdb";
    final String DB_USER = "root";
    final String DB_PASSWORD = "XXXXX";

    // Helper to escape HTML special characters in values shown on
the page
    String esc(String s) {
        if (s == null) return "";
        return s.replace("&", "&amp;").replace("<",
"&lt;").replace(">", "&gt;").replace("\"", "&quot;");
    }
%>
<%
    request.setCharacterEncoding("UTF-8");
```

```
    String action = request.getParameter("action");
    String message = "";
    boolean editing = false;
    String eId = "", eName = "", eDept = "", eSalary = "";
    java.util.List<String[]> rows = new
java.util.ArrayList<String[]>();

    try {
        Class.forName("com.mysql.cj.jdbc.Driver");
        try (Connection conn = DriverManager.getConnection(DB_URL,
DB_USER, DB_PASSWORD)) {

            if ("add".equals(action)) {
                // CREATE
                try (PreparedStatement ps = conn.prepareStatement(
                        "INSERT INTO jsp_employees (emp_id, name,
department, salary) VALUES (?, ?, ?, ?)")) {
                    ps.setInt(1,
Integer.parseInt(request.getParameter("empId").trim()));
                    ps.setString(2, request.getParameter("name"));
                    ps.setString(3,
request.getParameter("department"));
                    ps.setDouble(4,
Double.parseDouble(request.getParameter("salary").trim()));
                    ps.executeUpdate();
                    message = "Employee added successfully";
                }
            } else if ("update".equals(action)) {
                // UPDATE
                try (PreparedStatement ps = conn.prepareStatement(
                        "UPDATE jsp_employees SET name = ?,
department = ?, salary = ? WHERE emp_id = ?")) {
                    ps.setString(1, request.getParameter("name"));
                    ps.setString(2,
request.getParameter("department"));
                    ps.setDouble(3,
Double.parseDouble(request.getParameter("salary").trim()));
                    ps.setInt(4,
Integer.parseInt(request.getParameter("empId").trim()));
                    ps.executeUpdate();
                    message = "Employee updated successfully";
                }
            } else if ("delete".equals(action)) {
                // DELETE
                try (PreparedStatement ps =
conn.prepareStatement("DELETE FROM jsp_employees WHERE emp_id =
?")) {
                    ps.setInt(1,
Integer.parseInt(request.getParameter("id").trim()));
```

```
                    int deleted = ps.executeUpdate();
                    message = (deleted > 0) ? "Employee deleted
successfully" : "No employee found with that ID";
                }
            } else if ("edit".equals(action)) {
                // Load one record into the form for editing
                try (PreparedStatement ps = conn.prepareStatement(
                        "SELECT emp_id, name, department, salary
FROM jsp_employees WHERE emp_id = ?")) {
                    ps.setInt(1,
Integer.parseInt(request.getParameter("id").trim()));
                    try (ResultSet rs = ps.executeQuery()) {
                        if (rs.next()) {
                            editing = true;
                            eId =
String.valueOf(rs.getInt("emp_id"));
                            eName  =  rs.getString("name");
                            eDept = rs.getString("department");
                            eSalary =
String.valueOf(rs.getDouble("salary"));
                        }
                    }
                }
            }

            // READ - load all employees for the table
            try (Statement st = conn.createStatement();
                 ResultSet rs = st.executeQuery(
                         "SELECT emp_id, name, department, salary
FROM jsp_employees ORDER BY emp_id")) {
                while (rs.next()) {
                    rows.add(new String[]{
                            String.valueOf(rs.getInt("emp_id")),
                            rs.getString("name"),
                            rs.getString("department"),
                            String.valueOf(rs.getDouble("salary"))
                    });
                }
            }
        }
    } catch (Exception e) {
        message = "Error: " + e.getMessage();
    }
%>
<!DOCTYPE html>
<html>
<head>
    <title>Employee Management</title>
    <style>
```

```
        table, th, td { border: 1px solid #888; border-collapse:
collapse; padding: 6px 12px; }
    </style>
</head>
<body>
    <h2>Employee Management (CRUD)</h2>

    <% if (!message.isEmpty()) { %>
        <p><b><%= esc(message) %></b></p>
    <% } %>

    <form method="post" action="employee.jsp">
        <input type="hidden" name="action" value="<%= editing ?
"update" : "add" %>">
        Employee ID: <input type="text" name="empId" value="<%=
esc(eId) %>" <%= editing ? "readonly" : "" %> required><br><br>
        Name: <input type="text" name="name" value="<%= esc(eName)
%>" required><br><br>
        Department: <input type="text" name="department"
value="<%= esc(eDept) %>" required><br><br>
        Salary: <input type="text" name="salary" value="<%=
esc(eSalary) %>" required><br><br>
        <input type="submit" value="<%= editing ? "Update
Employee" : "Add Employee" %>">
        <% if (editing) { %> <a href="employee.jsp">Cancel</a> <%
} %>
    </form>

    <h3>Employee List</h3>
    <table>
        <tr><th>ID</th><th>Name</th><th>Department</th><th>Salary<
/th><th>Actions</th></tr>
        <% if (rows.isEmpty()) { %>
            <tr><td colspan="5">No records found</td></tr>
        <% } %>
        <% for (String[] r : rows) { %>
        <tr>
            <td><%= esc(r[0]) %></td>
            <td><%= esc(r[1]) %></td>
            <td><%= esc(r[2]) %></td>
            <td><%= esc(r[3]) %></td>
            <td>
                <a href="employee.jsp?action=edit&amp;id=<%=
esc(r[0]) %>">Edit</a>
                <form method="post" action="employee.jsp"
style="display:inline"
                      onsubmit="return confirm('Delete this
employee?');">
```

```
                    <input type="hidden" name="action"
value="delete">
                    <input type="hidden" name="id" value="<%=
esc(r[0]) %>">
                    <input type="submit" value="Delete">
                </form>
            </td>
        </tr>
        <% } %>
    </table>
</body>
</html>
```

2. Create a JSP-based application to manage patient records using CRUD operations. Include fields such as patient ID, name, age, disease, and treatment status.

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
pageEncoding="UTF-8" %>
<%@ page import="java.sql.*" %>
<%!
    // Database settings - change the password to match your MySQL
setup
    final String DB_URL = "jdbc:mysql://localhost:3306/testdb";
    final String DB_USER = "root";
    final String DB_PASSWORD = "XXXXX";

    // Helper to escape HTML special characters in values shown on
the page
    String esc(String s) {
        if (s == null) return "";
        return s.replace("&", "&amp;").replace("<",
"&lt;").replace(">", "&gt;").replace("\"", "&quot;");
    }
%>
<%
    request.setCharacterEncoding("UTF-8");
    String action = request.getParameter("action");
    String message = "";
    boolean editing = false;
    String pId = "", pName = "", pAge = "", pDisease = "", pStatus
= "";
    java.util.List<String[]> rows = new
java.util.ArrayList<String[]>();
```

```
    try {
        Class.forName("com.mysql.cj.jdbc.Driver");
        try (Connection conn = DriverManager.getConnection(DB_URL,
DB_USER, DB_PASSWORD)) {

            if ("add".equals(action)) {
                // CREATE
                try (PreparedStatement ps = conn.prepareStatement(
                        "INSERT INTO jsp_patients (patient_id,
name, age, disease, treatment_status) VALUES (?, ?, ?, ?, ?)")) {
                    ps.setInt(1,
Integer.parseInt(request.getParameter("patientId").trim()));
                    ps.setString(2, request.getParameter("name"));
                    ps.setInt(3,
Integer.parseInt(request.getParameter("age").trim()));
                    ps.setString(4,
request.getParameter("disease"));
                    ps.setString(5,
request.getParameter("status"));
                    ps.executeUpdate();
                    message = "Patient record added successfully";
                }
            } else if ("update".equals(action)) {
                // UPDATE
                try (PreparedStatement ps = conn.prepareStatement(
                        "UPDATE jsp_patients SET name = ?, age =
?, disease = ?, treatment_status = ? WHERE patient_id = ?")) {
                    ps.setString(1, request.getParameter("name"));
                    ps.setInt(2,
Integer.parseInt(request.getParameter("age").trim()));
                    ps.setString(3,
request.getParameter("disease"));
                    ps.setString(4,
request.getParameter("status"));
                    ps.setInt(5,
Integer.parseInt(request.getParameter("patientId").trim()));
                    ps.executeUpdate();
                    message = "Patient record updated
successfully";
                }
            } else if ("delete".equals(action)) {
                // DELETE
                try (PreparedStatement ps =
conn.prepareStatement("DELETE FROM jsp_patients WHERE patient_id =
?")) {
                    ps.setInt(1,
Integer.parseInt(request.getParameter("id").trim()));
                    int deleted = ps.executeUpdate();
```

```
                    message = (deleted > 0) ? "Patient record
deleted successfully" : "No patient found with that ID";
                }
            } else if ("edit".equals(action)) {
                // Load one record into the form for editing
                try (PreparedStatement ps = conn.prepareStatement(
                        "SELECT patient_id, name, age, disease,
treatment_status FROM jsp_patients WHERE patient_id = ?")) {
                    ps.setInt(1,
Integer.parseInt(request.getParameter("id").trim()));
                    try (ResultSet rs = ps.executeQuery()) {
                        if (rs.next()) {
                            editing = true;
                            pId =
String.valueOf(rs.getInt("patient_id"));
                            pName = rs.getString("name");
                            pAge =
String.valueOf(rs.getInt("age"));
                            pDisease = rs.getString("disease");
                            pStatus =
rs.getString("treatment_status");
                        }
                    }
                }
            }

            // READ - load all patients for the table
            try (Statement st = conn.createStatement();
                 ResultSet rs = st.executeQuery(
                         "SELECT patient_id, name, age, disease,
treatment_status FROM jsp_patients ORDER BY patient_id")) {
                while (rs.next()) {
                    rows.add(new String[]{
                            String.valueOf(rs.getInt("patient_id")
),
                            rs.getString("name"),
                            String.valueOf(rs.getInt("age")),
                            rs.getString("disease"),
                            rs.getString("treatment_status")
                    });
                }
            }
        }
    } catch (Exception e) {
        message = "Error: " + e.getMessage();
    }

    String[] statuses = {"Admitted", "Under Treatment",
"Recovered", "Discharged"};
```

```
%>
<!DOCTYPE html>
<html>
<head>
    <title>Patient Records</title>
    <style>
        table, th, td { border: 1px solid #888; border-collapse:
collapse; padding: 6px 12px; }
    </style>
</head>
<body>
    <h2>Patient Records Management (CRUD)</h2>

    <% if (!message.isEmpty()) { %>
        <p><b><%= esc(message) %></b></p>
    <% } %>

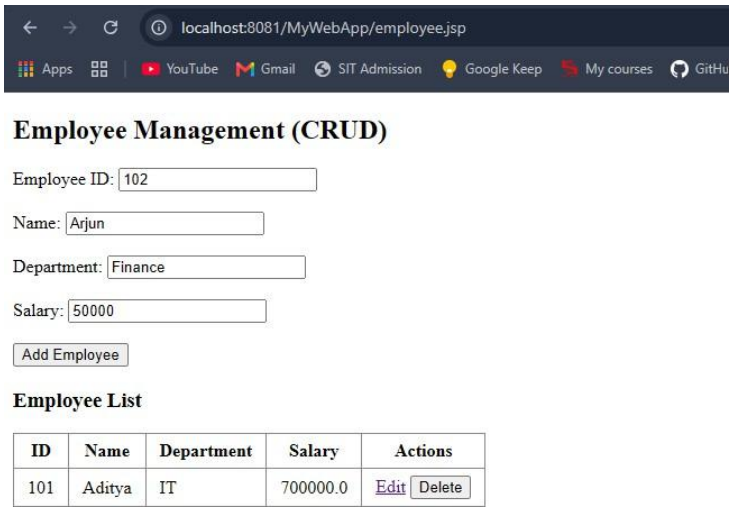
    <form method="post" action="patient.jsp">
        <input type="hidden" name="action" value="<%= editing ?
"update" : "add" %>">
        Patient ID: <input type="text" name="patientId" value="<%=
esc(pId) %>" <%= editing ? "readonly" : "" %> required><br><br>
        Name: <input type="text" name="name" value="<%= esc(pName)
%>" required><br><br>
        Age: <input type="text" name="age" value="<%= esc(pAge)
%>" required><br><br>
        Disease: <input type="text" name="disease" value="<%=
esc(pDisease) %>" required><br><br>
        Treatment Status:
        <select name="status">
            <% for (String s : statuses) { %>
                <option value="<%= s %>" <%= s.equals(pStatus) ?
"selected" : "" %>><%= s %></option>
            <% } %>
        </select><br><br>
        <input type="submit" value="<%= editing ? "Update Patient"
: "Add Patient" %>">
        <% if (editing) { %> <a href="patient.jsp">Cancel</a> <% }
%>
    </form>

    <h3>Patient List</h3>
    <table>
        <tr><th>ID</th><th>Name</th><th>Age</th><th>Disease</th><t
h>Treatment Status</th><th>Actions</th></tr>
        <% if (rows.isEmpty()) { %>
            <tr><td colspan="6">No records found</td></tr>
        <% } %>
        <% for (String[] r : rows) { %>
```

```
        <tr>
            <td><%= esc(r[0]) %></td>
            <td><%= esc(r[1]) %></td>
            <td><%= esc(r[2]) %></td>
            <td><%= esc(r[3]) %></td>
            <td><%= esc(r[4]) %></td>
            <td>
                <a href="patient.jsp?action=edit&amp;id=<%=
esc(r[0]) %>">Edit</a>
                <form method="post" action="patient.jsp"
style="display:inline"
                      onsubmit="return confirm('Delete this
patient record?');">
                    <input type="hidden" name="action"
value="delete">
                    <input type="hidden" name="id" value="<%=
esc(r[0]) %>">
                    <input type="submit" value="Delete">
                </form>
            </td>
        </tr>
        <% } %>
    </table>
</body>
</html>
```

Output

1)

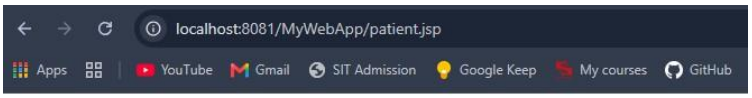


```
12
13 •  select * from jsp_employees;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

emp_id	name	department	salary
101	Aditya	IT	700000
102	Arjun	Finance	50000
NULL	NULL	NULL	NULL

2)



Patient Records Management (CRUD)

Patient record added successfully

Patient ID:

Name:

Age:

Disease:

Treatment Status: Admitted

Add Patient

Patient List

ID	Name	Age	Disease	Treatment Status	Actions
101	Rohit	23	Viral Fever	Under Treatment	Edit Delete
102	Rahul	30	Diabetes	Admitted	Edit Delete
103	Priya	35	Typhoid	Recovered	Edit Delete

```
13 •  select * from jsp_patients;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

patient_id	name	age	disease	treatment_status
101	Rohit	23	Viral Fever	Under Treatment
102	Rahul	30	Diabetes	Admitted
103	Priya	35	Typhoid	Recovered
NULL	NULL	NULL	NULL	NULL

Conclusion

A JSP CRUD application connects HTML forms to a database through JDBC. Create, update and delete are done with INSERT, UPDATE and DELETE statements run by a PreparedStatement, and read is a SELECT whose ResultSet is displayed as a table. Using PreparedStatement keeps user input separate from the SQL and so prevents SQL injection, and escaping the values when they are displayed prevents unwanted HTML from being injected into the page. Together these operations form the core of employee, patient, shopping and banking systems.